LAKSHYA JAIN

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EDUCATION

Indian Institute of Technology Jodhpur 2023 – 2027
B.Tech. in Computer Science and Engineering *CGPA: 8.97/10.0*

R.S. Memorial International School 2020 – 2022
Senior Secondary (CBSE) *93.8%*

St. Anne's Sr. Sec. School 2012 – 2020
Secondary (CBSE) *94.6%*

EXPERIENCE

Quant Developer Intern May 2026 – Jul 2026
26 Miles Capital – Quant Research Platform *On-site, India*

- Built an end-to-end quant research platform consolidating the full research workflow — data, backtesting, and workbench — across **25 modules** (**64** data models) with auth/ACL, credits, RunBT orchestration, and S3 workspaces; sole owner from architecture (**14 ADRs**) to production.
- Engineered multi-tenant isolated runtimes on **AWS ECS** with STS-scoped S3 access and idle-reclaim (**74%** sessions auto-reclaimed, p50 start **21s**); load-tested to **500 concurrent users** at p95 **19ms** with zero errors and projected **27%** AWS cost reduction.
- Established engineering rigor with **C4-style** architecture documentation (∼**91** active docs) and a KPI measurement pipeline tracking runtime, security, and test metrics (**65.1%** test coverage, **76/100** auth security score, ∼**11ms** avg API p95); layered JWT auth, a central canAccess/requireRole ACL, and dual-OS-user isolation to keep firm credentials out of researcher runtimes.

Founder & Lead Engineer Aug 2025 – Mar 2026
libr *– Censorship-Resilient Social Network* *Website*

- Founded and architected ***libr***, a censorship-resilient decentralized social network launched into a week-long public beta; attracted **250+ landing page visitors**, **570+ page views**, **100+ downloads**, and **1,000+ user posts**.
- Designed a 3-tier system with clients, database nodes, and moderators; built core services in **Go** with **Docker**, **PostgreSQL**, **Kademlia DHT** for content routing, and cryptographic certificates for tamper-evident storage.
- Implemented replicated **DHT**-based storage with partial immutability and **O(log n)** lookup, and designed a quorum-backed moderation layer, backed by risk analysis covering **20+** identified risks, to preserve content integrity across a distributed node set.

Domain Lead – Adobe Problem Statement Oct 2025 – Dec 2025
Inter IIT Tech Meet 14.0 *Github*

- Led a cross-functional team to develop an AI-powered Flutter image editing app, integrating on-device **YOLO** segmentation for real-time object masking and a modular client–service architecture with edge inference **under a 200 MB** device footprint.
- Developed isolated **Docker** AI compute containers on edge servers, orchestrating parallel editing pipelines (inpaint, erase, move, compositing) with **sub-30s end-to-end latency**; optimized the image editing pipeline for low-latency mobile interactions.
- Directed integration of a multi-model AI pipeline – **DDIM**-based latent move, YOLO-guided **Flux** inpainting, **U-Net** attention-based erase, and an instruction-tuned **Qwen2.5-3B**/Flux-Kontext compositing mode – balancing on-device and edge-server inference across the feature set.

SDE Intern – Android Development Dec 2024 – Apr 2025
PMStack *Playstore*

- Engineered **Pinnzo**, a Flutter-based Android application for organizing and summarizing reading materials for working professionals, deployed to the **Google Play Store** with Firebase real-time data synchronization.
- Partnered with startup founders to conceive and develop product features, reducing production time by **30%** through implementing **Agile** methodologies and **2-week** sprint cycles.
- Instrumented **Google Analytics** and **Crashlytics** to triage **163** critical bugs across Android versions, decreasing crash rates by **40%** and improving app load speed by **25%** to acquire **120+** beta users within the first month.

PROJECTS

Mini-K8ts – Container Orchestrator *2026*

Go, Docker, PostgreSQL, Control Plane, Prometheus, Grafana Github

- Developed a lightweight container orchestrator targeting edge environments with a **<50 MB control-plane RAM**, **25 MB binary**, and **<1 sec startup**; uses Go, Docker, and PostgreSQL as the ACID state store.
- Built a reliability-aware, multi-stage scheduler with shadow resource tracking; sustains **220+ tasks/sec** throughput, **97.8–100% scheduling efficiency**, converging 1000 tasks in **4.2 sec** and 2000 tasks in **9.1 sec**.
- Hardened the system for production with token-based RBAC auth, per-client rate limiting, optional mTLS, and Postgres-backed leader election for HA, plus offline-tolerant edge workers validated by a **50-node/1,000-task** scale-test suite alongside Prometheus/Grafana monitoring.

FileNest – Multi-Modal P2P Search Engine *Summer 2025*

Go, Python, React, P2P (libp2p-kad-dht), CLIP, FAISS, Redis Github

- Mentored a **hierarchical distributed tagging system** combining semantic embeddings with P2P routing (Kademlia DHT), enabling multi-modal search across text, PDFs, images, and video on a peer-to-peer network.
- Designed an AI/ML pipeline integrating **BERT, Sentence-Transformers, and CLIP** for cross-modal understanding; targeted **15–20 files/sec** indexing throughput and sub-100ms query latency across 1000+ concurrent users.
- Set up the project's **CI/CD** workflows and modular monorepo architecture (AI, backend, network, frontend) across the Go/Python/React stack, integrating Redis caching with SQLite/PostgreSQL metadata storage during **Summer RAID 2025** at IIT Jodhpur.

GitCury – AI-Powered Git Automation CLI *Apr. 2025*

Go, Cobra CLI, Redis, NLP, Machine Learning Github

- Created an intelligent Git workflow automation tool that cut commit time **from 17 minutes to 32 seconds (97% faster)**, handling 298 files with a 4 MB memory footprint via parallel AI-powered commit generation.
- Reduced external API calls by **90%** through semantic file clustering algorithms and batched operations; semantic clustering achieves **3.2× faster** processing than traditional approaches in multi-repository environments.
- Architected a dependency-injected, interface-based codebase with a concurrent worker pool for multi-repo batch operations, exposing **3** selectable clustering strategies (directory, pattern, semantic) and shipping via Homebrew, Scoop, Docker, and **6** OS/architecture binaries.

2nd Swap – Secure, Intelligent Second-Hand Trading *Feb. 2025 – Mar. 2025*

Node.js, Express.js, MongoDB, Google OAuth 2.0, Cloudinary Website

- Architected and deployed the complete backend within **1.5 weeks**, implementing **12+** RESTful API endpoints with **99.8% uptime** for this startup.
- Developed an AI-powered verification system that generates personalized, item-specific questions and automates the validation process, reducing manual intervention by **20%** while processing over **30+** daily queries.
- Implemented a secure object verification mechanism using **OCR-based code recognition** to uniquely identify and validate items, ensuring accurate verification for all listings.

NIDS – Network Intrusion Detection System *2025*

Python, FastAPI, Scapy, Machine Learning (Random Forest) Github

- Built an open-source network intrusion detection system in Python with a **FastAPI** backend that captures live traffic via **Scapy** and classifies packets as normal or anomalous in real time.
- Implemented a custom **Random Forest** classifier – bootstrapped feature-subspace sampling with information-gain decision trees – trained on **14** engineered packet and connection features, including service/host tracking and statistical traffic patterns.
- Exposed the detection pipeline through REST endpoints (health check, capture-and-classify) with configurable detection parameters, and documented a promiscuous-mode Linux deployment workflow for packet capture.

C3P – Collaborative Content Platform *Dec. 2024*

React.js, Express.js, Node.js, MongoDB, Social Media APIs Github

- Constructed a full-stack platform enabling collaborative content creation with **8** user roles and streamlined publishing to **4** major social media platforms, reducing publishing time by **65%**.
- Devised role-based access control mechanisms and content ranking algorithms driven by **6** key engagement metrics, increasing user retention by **22%**.
- Orchestrated automated publishing workflows and revenue-sharing schemes for monetized content, processing **$1200+** in transactions with **0** payment errors.

RealmWeaver – Dynamic Game Environment *Sep. 2024 – Nov. 2024*

C++, Gemini API, Data Structures and Algorithms Github

- Engineered a dynamic game world with automated, AI-driven content generation, allowing the creation of **NPCs, quests, and locations in real-time**, ensuring a constantly evolving and personalized gameplay experience.
- Optimized content generation by integrating **multiple round-robin API keys**, exponentially reducing creation time, speeding up the overall content creation from **5 mins** to **less than 1 min** for 3 keys.

- Developed a centralized content management system that reduced the manual creation of NPCs, quests, and storylines by **100x**, enabling seamless and efficient world-building with minimal human input.

GenCLI – Semantic File Retrieval Tool *May 2024 – Jun. 2024*
Go, Cobra, Gemini API Github

- Optimized file indexing performance from **19.2 seconds/file** to **1.2 seconds/file** by implementing batch processing and parallelization techniques, significantly improving efficiency.
- Enhanced user experience with **7+ customizable themes**, providing a visually appealing and easy-to-use interface tailored to diverse user preferences.
- Achieved **sub-second search response time**, enabling fast retrieval of indexed content across **50+ files** with advanced AI-driven semantic search capabilities.

The Vindication – Text-Based Adventure Game *Apr. 2024 – May. 2024*
C, JSON, ASCII Graphics Github

- Crafted an open-world adventure game featuring **27 quests** and **7 mini-games** using modular JSON-based data structures, achieving **10+ hours** of gameplay.
- Programmed **120+** ASCII terminal graphics elements and integrated **15** in-game audio tracks to enhance user immersion, increasing average play session by **45%**.
- Formulated save/load functionality via JSON parsing to support **unlimited** unique game states and **25+** customization options with **100%** data integrity.

Image Filtration Program – C-Based Image Processing *Mar. 2024*
C, GCC, BMP File Format Github

- Programmed **6** image filters (Grayscale, Blur, Sharpen, Sobel Edge Detection, Reflect, Negative) as an ICS minor project in C.
- Implemented reading and writing of 24-bit BMP images, saving each filtered result to a new file.

Travel Companion App – Your Adventure, Simplified *Jan. 2024 – Mar. 2024*
Flutter, Firebase, Figma Playstore

- Developed a feature-rich mobile application exclusively for our college community, achieving **360+ registered users** and maintaining a **5-star rating** on user feedback.
- Incorporated **7 core functionalities** including trip sharing, advanced filtering, and seamless search, enhancing the overall user experience.
- Integrated a real-time user request system using Firebase, delivering **sub-1s** response times to ensure smooth coordination of trip join requests among **10+ concurrent users**.

Technical Skills

Languages: Go, C, C++, Java, Python, JavaScript, TypeScript, Dart
Frameworks & Libraries: Flutter, React.js, Node.js, Express.js, Firebase, Cobra, Socket.io, Multer, Cloudinary, REST APIs
Databases & Storage: PostgreSQL, MongoDB, Redis, Firestore, FAISS / Vector DBs
DevOps, Cloud & Tools: Git, Docker, Kubernetes, Container Orchestration, Prometheus, Grafana, CI/CD (GitHub Actions), AWS (ECS/EC2/S3/STS), Postman, Trello
Systems & Software Concepts: System Design, Distributed Systems, Microservices, P2P Networking, Multi-threading & Concurrency, Computer Networks, Cryptography, Risk Analysis, Linux
AI / NLP Concepts: Vectorization, Embeddings, Semantic Search, AI-driven Querying, GPT Integration, Chatbot Development, Computer Vision (CLIP, YOLO), Fine-tuning
Soft Skills: Critical Thinking, Teamwork, Problem Solving, Communication, Leadership, Time Management
Other: Google OAuth 2.0, Responsive Design, Unit Testing, Debugging, OCR

Coursework

Computer Science: Data Structures & Algorithms, Software Engineering (A), Database Management Systems (A), Operating Systems (A-), Computer Organization and Architecture (A), Embedded Systems (A), Computer Networks (A-), Cyber Security (A-), Blockchain (A), Virtualization and Cloud Computing (A-), Introduction to Computer Science (A*), Pattern Recognition and Machine Learning
Mathematics: Mathematics for Computing (A), Mathematics-I: Calculus (A), Mathematics-II: Linear Algebra (A-), Probability, Statistics and Stochastic Processes (A-), Foundations of Quantum Information (B)

Achievements

- **Ranked 6th among 23 IITs** at **Inter IIT Tech Meet 14.0** for the Adobe Problem Statement.
- Serving as **Head/Coordinator** of **Devlup Labs**, the open-source community at IIT Jodhpur, mentoring **20+** contributors.
- Delivered a **System Design** session to **100+** students covering scalable backend architecture and distributed systems.
- Led the team to a **Gold Medal** in **Bug Bounty '25**, an open-source competition, by resolving critical bugs in existing projects.

- Secured **Rs. 80,000+** in industry sponsorships as Assistant Head of **Prometeo '25** through corporate partnerships.
- Increased **E-Cell Vandre '25** profit by **500%** to **Rs. 2,28,153** through resource optimization and strategic planning.
- Secured **1st place** in **CADastra** at **Prometeo '25**, IIT Jodhpur's Tech Fest, for innovative CAD design solutions.
- Won **1st place** in **Pitch It** at **Initio '23**, E-Cell IIT Jodhpur's startup competition, with a compelling business proposal.

Positions of Responsibility

Head/Coordinator, DevlUp Labs, IIT Jodhpur — *Mar. 2025 – Present*
Core Member, Quant Club, IIT Jodhpur — *May. 2024 – Present*
Associate, Product Club, IIT Jodhpur — *May. 2024 – Present*
Senior Executive, E-cell, IIT Jodhpur — *Jun. 2024 – Present*

Extracurricular

Student Guide, Student Wellbeing Committee, IIT Jodhpur — *Jun. 2024 – Apr. 2025*
Assistant Head, Prometeo X Econclave (Tech Fest), IIT Jodhpur — *2025*
Volunteer, Varshas (Sports Fest), IIT Jodhpur — *2024*